

Tristan Dyson

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EDUCATION

UON (University Of Newcastle)

Bachelor of Data Science / Bachelor of Computer Science
Major in Artificial Intelligence
WAM: 92; GPA: 7.0 / 7.0

Newcastle, NSW
2025 - Present
Expected 2028

Academic Excellence Scholarship Recognising Outstanding Performance in Undergraduate Studies

Scholarship awarded to top 32 undergraduate students across all programs based on WAM, valued at \$10k

High Performing Students Program - BUSN 1001: Globalisation and International Management

Scheme providing high school students access to university level courses

2022 @ UON

EXPERIENCE

Software Engineer | NUbots Robotics Research Group | C++, Typescript

Worked on the software of humanoid soccer robots that represent Australia internationally

September 2025 - Present

github.com/NUbots

- Reduced misdetections in the computer vision pipeline by combining clusters of detections on the same ball.
- Implemented robot radius detector and added footprint visualisation into the web visualiser for easy debugging.
- Collaborated in a complex 100k+ line codebase alongside a large team using Jira and Git for agile development.

PROJECTS

Benchmark Collection | C++

Benchmarks, verified through statistical testing such as confidence intervals and ANOVAs

January 2026 - Present

github.com/dargem/benchmark_fun

- Designed microbenchmarks for memory layouts (SOA vs AOS), SIMD, branch prediction, and RNG performance
- Showed results were reliable through statistical testing such as confidence intervals and ANOVA with post-hocs
- Achieved up to 10x speedups with SOA + SIMD on uint8_t data and up to ~7.5x gains from parallel execution policies
- Designed multiple lock-free MPMC queue implementations, finding from 2x improvements to 30x slow downs vs a mutex baseline
- Discovered intricacies like long double SIMD gains disappearing due to lack of AVX2/AVX-512 support for 80-bit floats

SIMD Accelerated RNG | C++

Header only library providing a SIMD accelerated implementation of Xoroshiro64*

May 2026 - June 2026

github.com/dargem/Xoroshiro64StarSIMD

- 10.2 GB/s generation speeds, achieves a 7x speedup over the original scalar algorithm and is 23x faster than MT19937
- Compile time ISA detection and dispatch (AVX/AVX2/AVX512) via type traits, highly extensible to new instruction sets
- Zero-overhead scalar wrapper, variadic templates and compile-time optional fields mean callers pay only for the types they use.

Graphics Engine | C++, Java

Procedurally generated ASCII background art using a custom graphics renderer and ECS

November 2025 - Present

github.com/dargem/TUI-Art

- Creating a heavily templated Entity Component System enabling cache-efficient, data-oriented architecture
- Created an efficient graphics rendering pipeline for rasterizing meshes
- Implemented smart diffing engine exploiting viewport movements and line prints to reduce character output by 23%

MMO Combat Simulator | C++, Java

Technologies: Springboot, React

August 2024 - December 2025

github.com/dargem/Vikings-Rise-Simulator

- Deployed a cloud hosted MMO combat simulator with ~200 monthly users
- Derived combat formulas through regression analysis using in-game sampling
- Reconstructed backend combat system with faithful modelling through monte carlo simulations

AWARDS & SKILLS

Placed on the **UON College Commendation list** for 2025

Languages: C++, Java, Python, R, SQL

Tools / Frameworks: Git, Jira, spaCy, LangGraph, Pandas, Eigen, Neo4j

Extracurriculars: Competitive Programming, competing in SPAR (South Pacific Algorithmic Rounds)

Interests: Modern C++, operating systems, performant code, computer graphics and data-oriented design